HW 1 — Faraday Rotation & Jackson 7.3

How to read this: black = write this down · **red = the answer** · *teal = explanation, no need to copy* · grey box = optional, skip it

Every starting equation carries a tag like (Lec 3, slide 2) *telling you which slide it comes from.*

Problem 1 — Faraday rotation: θ(z)

In the Faraday-rotation example the medium has index $\mathbf{n_L}$ for left-circular and $\mathbf{n_R}$ for right-circular polarization. At z = 0 the wave is linearly polarized along $\hat{x}$, i.e. $\mathbf{E}(z=0, t) = \hat{x}\, E\, e^{-i\omega t}$. Using $\tan\theta = E_y/E_x$ (both real), find how fast the plane of polarization rotates with z.

ANSWER

$$\theta(z) = -\frac{(k_L - k_R)z}{2} = \frac{\omega}{2c}(n_R - n_L)z$$

$$\frac{d\theta}{dz} = \frac{\omega}{2c}(n_R - n_L) = \textbf{a constant} \quad [\text{rad/m}]$$

θ grows ***linearly*** *in z, so the rotation rate never changes along the path. The wave stays* ***linearly polarized everywhere*** *— only its direction turns.*

Step 0 — Where does the starting equation come from?

This is the part you have to know before you can write anything. It is all on Lecture 3, slides 2–3.

Why write the wave as two circular waves?

Because the problem hands you ***two*** *indices, n_L and n_R. An index belongs to a wave that travels without changing shape. So in this medium the only two waves that propagate cleanly are left- and right-circular, each with its own k = ωn/c. A linearly polarized wave is* ***not*** *a natural wave here — it has to be built out of those two.*

$$\vec{E}(z,t) = \vec{E}_L\, e^{i(k_L z - \omega t)} + \vec{E}_R\, e^{i(k_R z - \omega t)}$$ (Lec 3, slide 2)

$$\vec{E}_L = \mathscr{E}_L \hat{e}_+, \quad \vec{E}_R = \mathscr{E}_R \hat{e}_-, \quad \hat{e}_\pm = \frac{1}{\sqrt{2}}(\hat{x} \pm i\hat{y})$$ (Lec 2, complex unit vectors)

$\mathscr{E}_L$ and $\mathscr{E}_R$ are two unknown numbers. Everything else above is known. Finding them is the whole of Step 0.

How do we fix $\mathcal{E}_L$ and $\mathcal{E}_R$?

Use the one piece of information the problem gives: what the wave looks like at z = 0. Set z = 0 in the equation above; the two exponentials both become $e^{-i\omega t}$ and cancel against the right-hand side.

$$\vec{E}(0,t) = \left(\mathcal{E}_L \hat{e}_+ + \mathcal{E}_R \hat{e}_- \right) e^{-i\omega t} = \hat{x} E e^{-i\omega t} \quad \text{(given)}$$

$$\Rightarrow \quad \frac{\mathcal{E}_L}{\sqrt{2}} (\hat{x} + i\hat{y}) + \frac{\mathcal{E}_R}{\sqrt{2}} (\hat{x} - i\hat{y}) = \hat{x} E$$

compare $\hat{y}$: $\quad \frac{i}{\sqrt{2}} \left(\mathcal{E}_L - \mathcal{E}_R \right) = 0 \Rightarrow \mathcal{E}_L = \mathcal{E}_R$

compare $\hat{x}$: $\quad \frac{1}{\sqrt{2}} \left(\mathcal{E}_L + \mathcal{E}_R \right) = E \Rightarrow \mathcal{E}_L = \mathcal{E}_R = \frac{E}{\sqrt{2}}$

STARTING EQUATION — everything from here on is worked out from this one line:

$$\vec{E}(z,t) = \frac{E}{\sqrt{2}} \hat{e}_+ e^{i(k_L z - \omega t)} + \frac{E}{\sqrt{2}} \hat{e}_- e^{i(k_R z - \omega t)}$$

= the last line of Lecture 3, slide 3

Step 1 — Split it into x and y components

Put $\hat{e}_\pm = (\hat{x} \pm i\hat{y})/\sqrt{2}$ into the starting equation and collect the $\hat{x}$ terms and the $\hat{y}$ terms.

$$E_x = \frac{E}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \left(e^{ik_L z} + e^{ik_R z} \right) e^{-i\omega t} = \frac{E}{2} \left(e^{ik_L z} + e^{ik_R z} \right) e^{-i\omega t}$$

$$E_y = \frac{E}{2} \left(i e^{ik_L z} - i e^{ik_R z} \right) e^{-i\omega t} = i \frac{E}{2} \left(e^{ik_L z} - e^{ik_R z} \right) e^{-i\omega t}$$

Step 2 — Combine the two exponentials

Standard trick: pull out the average phase, and what is left is a cos and a sin.

Let $\bar{k} = \frac{k_L + k_R}{2}$ (average), $\Delta k = k_L - k_R$ (difference). Then $k_L = \bar{k} + \Delta k/2$ and $k_R = \bar{k} - \Delta k/2$, so

$$e^{ik_L z} \pm e^{ik_R z} = e^{i\bar{k}z}\left(e^{i\Delta kz/2} \pm e^{-i\Delta kz/2}\right) = 2\,e^{i\bar{k}z}\cos\frac{\Delta kz}{2} \quad \text{or} \quad 2i\,e^{i\bar{k}z}\sin\frac{\Delta kz}{2}$$

$$\Rightarrow \quad E_x = E\,e^{i(\bar{k}z-\omega t)}\cos\frac{\Delta kz}{2}, \qquad E_y = -E\,e^{i(\bar{k}z-\omega t)}\sin\frac{\Delta kz}{2}$$

(E_y: the i from Step 1 times the $2i$ from the sine gives $i\cdot i = -1$.) Both components carry the ***same*** *factor $E\,e^{i(\bar{k}z-\omega t)}$; divide it out and only cos and −sin remain, which are real. That is the "E_x, E_y both real" the problem mentions.*

Step 3 — Take the ratio and read off θ

The common factor cancels in the ratio, which is why θ is a real angle.

$$\tan\theta(z) = \frac{E_y}{E_x} = \frac{-\sin(\Delta kz/2)}{\cos(\Delta kz/2)} = \tan\left(-\frac{\Delta kz}{2}\right) \quad \Rightarrow \quad \theta(z) = -\frac{\Delta kz}{2}$$

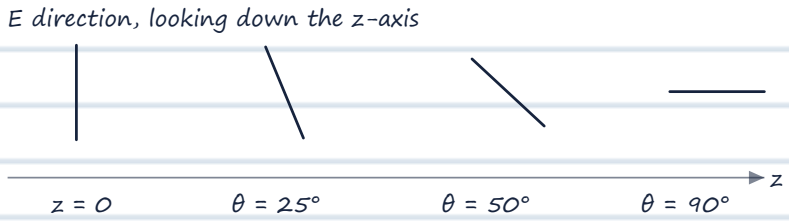
with $k = \frac{\omega n}{c}$ (Lec 2): $\Delta k = \frac{\omega}{c}(n_L - n_R)$

FINAL ANSWER

$$\theta(z) = \frac{\omega}{2c}(n_R - n_L)z \qquad \frac{d\theta}{dz} = \frac{\omega}{2c}(n_R - n_L) = \frac{\pi}{\lambda_0}(n_R - n_L)$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

Over a cell of length d the plane has turned by $\beta = (\omega/2c)(n_R - n_L)d$ — the angle β on the "Application of Faraday Rotation" slide. In a magnetized medium $(n_L - n_R) \propto B$, giving $\theta = VBd$ with V the Verdet constant.



Equal steps in z give equal steps in θ.

OPTIONAL **Check against the lecture.** The slide asked where the wave is polarized along $\hat{y}$, i.e. $E_x = 0$, so $\cos(\Delta kz/2) = 0 \Rightarrow (k_L - k_R)z = (2m+1)\pi$ — exactly the slide's answer. ✓

Problem 2 — Jackson 7.3

> **7.3** Two semi-infinite slabs of the same dielectric (index n, nonpermeable, lossless) are parallel, separated by an air gap (n = 1) of width d. A plane wave of frequency ω hits the gap from inside one slab at angle of incidence i. For polarization both parallel to and perpendicular to the plane of incidence: **(a)** find transmitted/incident power and reflected/incident power; **(b)** for i above the critical angle, sketch transmitted/incident power vs. d in units of the wavelength in the gap.

ANSWER

$\cos r = \sqrt{1-n^2\sin^2 i}$ (angle inside the gap), $\beta = \frac{\omega d}{c}\cos r$ (phase for one crossing of the gap).

(a) both polarizations:

$$T = \frac{1}{1 + F\sin^2\beta}, \quad R = 1 - T, \quad F_s = \frac{(n^2-1)^2}{4n^2\cos^2 i\cos^2 r}, \quad F_p = F_s[(n^2+1)\sin^2 i - 1]^2$$

(b) above critical angle: $\gamma = \sqrt{n^2\sin^2 i - 1}$, $\alpha = 2\pi\gamma d/\lambda$, and $\sin^2\beta \to -\sinh^2\alpha$:

$$T_s = \left[\, 1 + \frac{(n^2-1)^2}{4n^2\gamma^2\cos^2 i}\,\sinh^2\alpha\,\right]^{-1}, \quad T_p = \text{same} \times [(n^2+1)\sin^2 i - 1]^2$$

Step 0 — How do you even start this one?

Nothing new is needed — it is the lecture's single-interface problem used twice, plus bookkeeping for the bounces in between.

Recognise the shape of the problem

1. Two interfaces (slab→gap at $z = 0$, gap→slab at $z = d$).
2. At <u>each</u> one, the lecture already gives r and t (Lec 4, slides 3–4).
3. Inside the gap the wave bounces back and forth forever ⇒ add up a geometric series.
4. Above the critical angle the gap wave is evanescent (Lec 4, slide 8), so $\cos r$ goes imaginary — that is all part (b) is.

So the plan is: ***angles → one-interface coefficients → sum the bounces → square it.*** *Steps 1–4 below are exactly those four things.*

Step 1 — Angles: Snell twice

Same kinematics as Lecture 3 — do the angles before touching amplitudes.

slab → gap: $n \sin i = 1 \cdot \sin r \Rightarrow \sin r = n \sin i$, $\cos r = \sqrt{1 - n^2 \sin^2 i}$ (Lec 3, slide 12)

gap → slab: $1 \cdot \sin r = n \sin i' \Rightarrow i' = i$ *exit angle = entry angle*

critical angle: $\sin i_c = 1/n$

*Slab 3 is the **same medium** as slab 1 and the exit angle is the same, so the power ratio is just $T = |t|^2$ — the flux factor $(Z/Z')(\cos r/\cos i)$ from Lecture 4 equals 1 here.*

Step 2 — Reflection at one interface

Straight from the lecture's Fresnel equations, written for slab (n) → air (1), i.e. $Z = Z_0/n$ and $Z' = Z_0$

$$r_s = \frac{Z'\cos i - Z\cos r}{Z'\cos i + Z\cos r} = \frac{n\cos i - \cos r}{n\cos i + \cos r}, \quad r_p = \frac{Z\cos i - Z'\cos r}{Z\cos i + Z'\cos r} = \frac{\cos i - n\cos r}{\cos i + n\cos r}$$

(Lec 4, sl. 3–4)

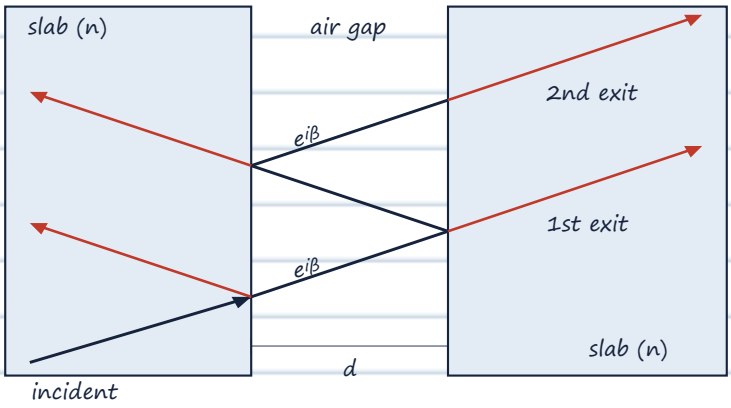
Going the other way (air → slab) swaps the two media in those same formulas:

$r_{21} = -r_{12}$ and $t_{12}t_{21} = 1 - r_{12}^2$ *swap $n \leftrightarrow 1$ and $i \leftrightarrow r$ to check*

Write r for r_{12} from here on. Everything below is identical for s and p — only the value of r differs.

Step 3 — Add up every bounce inside the gap

*Each crossing of the gap multiplies the amplitude by $e^{i\beta}$, where β is the phase gained by the z-component of **k**: $\beta = (\omega d/c)\cos r$.*



Each crossing costs $e^{i\beta}$; each extra round trip costs $r_{21}^2 e^{2i\beta}$.

$$t_{tot} = t_{12}t_{21}e^{i\beta}\left[\,1 + r^2e^{2i\beta} + r^4e^{4i\beta} + \dots\,\right] = \frac{(1-r^2)\,e^{i\beta}}{1 - r^2e^{2i\beta}} \qquad 1+x+x^2+\dots = 1/(1-x)$$

Step 4 — Square it to get the power

$$\left|\,1 - r^2e^{2i\beta}\,\right|^2 = 1 - 2r^2\cos 2\beta + r^4 = (1-r^2)^2 + 4r^2\sin^2\beta \qquad 1-\cos 2\beta = 2\sin^2\beta$$

$$\Rightarrow\quad T = \left|t_{tot}\right|^2 = \frac{(1-r^2)^2}{(1-r^2)^2+4r^2\sin^2\beta} = \frac{1}{1 + F\sin^2\beta}\,, \qquad F \equiv \frac{4r^2}{(1-r^2)^2}$$

Step 5 — Put in r_s and r_p → part (a)

Only F changes between the two polarizations.

$$1 - r_s^2 = \frac{(n\cos i+\cos r)^2 - (n\cos i-\cos r)^2}{(n\cos i+\cos r)^2} = \frac{4n\cos i\cos r}{(n\cos i+\cos r)^2}$$

$$\Rightarrow\quad F_s = \frac{4r_s^2}{(1-r_s^2)^2} = \frac{(n^2\cos^2 i - \cos^2 r)^2}{4n^2\cos^2 i\cos^2 r}$$

now $\cos^2 r = 1-n^2\sin^2 i$, so the numerator collapses: $n^2\cos^2 i - 1 + n^2\sin^2 i = n^2-1$

same for p: $\cos^2 i - n^2\cos^2 r = (n^2-1)[(n^2+1)\sin^2 i - 1]$

FINAL ANSWER — PART (a)

$$F_s = \frac{(n^2-1)^2}{4n^2\cos^2 i\cos^2 r}, \qquad F_p = \frac{(n^2-1)^2[(n^2+1)\sin^2 i - 1]^2}{4n^2\cos^2 i\cos^2 r}$$

$$T_{s,p} = [\,1 + F_{s,p}\sin^2\beta\,]^{-1}, \qquad R_{s,p} = 1 - T_{s,p}, \qquad \beta = (\omega d/c)\cos r$$

Step 6 — Above the critical angle → part (b)

Now $\sin r = n\sin i > 1$, so $\cos r$ cannot be a real number — it goes imaginary. This is the evanescent wave of Lecture 4.

$\cos r = i\gamma$, $\gamma = \sqrt{n^2\sin^2 i - 1}$ (Lec 4, slide 8) $\Rightarrow$ $\beta = i\alpha$, $\alpha = \frac{\omega d\gamma}{c} = 2\pi\gamma\,\frac{d}{\lambda}$

$\sin^2\beta = \sin^2(i\alpha) = -\sinh^2\alpha$ and $\cos^2 r = -\gamma^2$

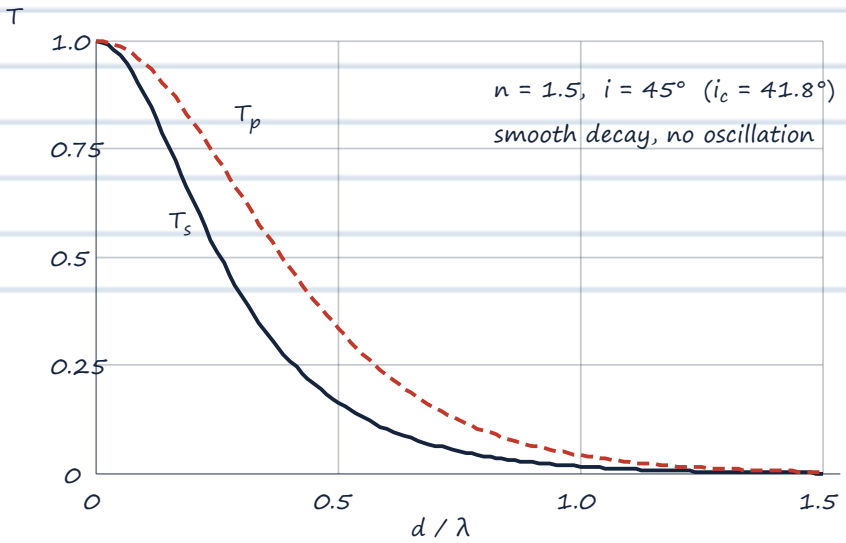
the two minus signs cancel inside $F\sin^2\beta$, so T stays real and positive.

FINAL ANSWER — PART (b)

$$T_s = [\,1 + \frac{(n^2-1)^2}{4n^2\gamma^2\cos^2 i}\,\sinh^2\alpha\,]^{-1}, \qquad T_p = [\,1 + \frac{(n^2-1)^2[(n^2+1)\sin^2 i - 1]^2}{4n^2\gamma^2\cos^2 i}\,\sinh^2\alpha\,]^{-1}$$

*The gap is vacuum so its wavelength is λ, and d/λ is exactly the axis Jackson wants. sinh is monotonic $\Rightarrow$ T falls smoothly from 1 to 0, **no oscillation**. For $d \gg \lambda$: $T \propto e^{-4\pi\gamma d/\lambda}$ (photon tunnelling).*

The sketch asked for in part (b)



Transmitted / incident power vs. gap width, above the critical angle.

Values used (n = 1.5, i = 45°): $d/\lambda = 0.1 \to T_s = 0.88$, $T_p = 0.95$; 0.25 → 0.51, 0.73; 0.5 → 0.16, 0.33; 1.0 → 0.017, 0.043.

■ END OF REQUIRED SOLUTION — everything below is optional ■

OPTIONAL **Three quick checks on part (a).**

- $d = 0 \Rightarrow \beta = 0 \Rightarrow T = 1$: the slabs touch, no interface. ✓
- $\beta = m\pi$, i.e. $d \cos r = m\lambda/2 \Rightarrow T = 1$ — Fabry–Pérot resonance. ✓
- $F_p = 0$ when $\tan i = 1/n$ — Brewster's angle, so $T_p = 1$ for every d. ✓